Orientation Materials for the PQC Hard Problem Colloquium

Your Name

Your Institution / Research Group

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Purpose of This Colloquium

- This colloquium is about the **hard problems** that underlie modern public-key cryptography and post-quantum cryptography.
- The goal is not only to know *which schemes exist*, but to understand:
  - what mathematical problem each scheme relies on,
  - what the best known attacks are,
  - how classical and quantum complexity differ,
  - and why some families remain viable while others have failed.
- In other words, the central object of study is the **security assumption itself**.

Guiding Question

Core Question

Which computational problems remain hard enough to support secure cryptography in the presence of quantum computers?

This question naturally divides our discussion into three layers:

1. the **mathematical hard problem**,
2. the **best known attacks**,
3. the **cryptosystems** built on top of that assumption.

Why Hard Problems Matter

- A cryptosystem is only as trustworthy as the computational problem beneath it.
- Cryptographic history shows that assumptions can fail in two very different ways:
  - through a major algorithmic breakthrough,
  - or through hidden algebraic structure that makes a scheme easier than expected.
- Post-quantum cryptography therefore requires a careful comparison of:
  - **classical hardness**,
  - **quantum hardness**,
  - **practical attack cost**,
  - and **structural fragility**.

The Classical Baseline

Before post-quantum cryptography, public-key cryptography was dominated by a small number of number-theoretic assumptions:

- **Integer Factorization** $\rightarrow$ RSA, Rabin
- **Discrete Logarithm in Finite Fields** $\rightarrow$ Diffie–Hellman, DSA, ElGamal
- **Elliptic Curve Discrete Logarithm** $\rightarrow$ ECDH, ECDSA, EdDSA

These problems were historically attractive because they enabled elegant and efficient cryptographic constructions.

Why the Classical Baseline Fails in the Quantum Model

- Classically, factorization and finite-field discrete logarithm admit only **sub-exponential** attacks in their best known forms.
- Elliptic-curve discrete logarithm is even stronger classically, with generic attacks of roughly $O(\sqrt{n})$.
- However, in the quantum setting, all three become vulnerable to **Shor's algorithm**.

Consequent Shift

The PQC transition is fundamentally the replacement of Shor-vulnerable assumptions with new assumptions for which no efficient quantum attacks are known.

A Historical Security Lesson

The classical world teaches an important lesson for PQC:

- security is not only about asymptotic hardness,
- it is also about the existence of **unexpected algorithmic structure**,
- and about whether the problem remains hard under the exact distribution used by the scheme.

This is why the colloquium must focus on **hard problems first**, not only on named standards or implementations.

The Main PQC Families

The major families discussed in post-quantum cryptography are:

1. **Lattice-based problems**
2. **Code-based problems**
3. **Hash-based assumptions**
4. **Multivariate polynomial problems**
5. **Isogeny-based problems**

Each family has a different profile with respect to:

- attack methodology,
- concrete efficiency,
- reduction theory,
- and quantum resistance.

Lattice Problems

The lattice family includes problems such as:

- **LWE** (Learning With Errors),
- **SIS** (Short Integer Solution),
- **SVP** (Shortest Vector Problem),
- **CVP** (Closest Vector Problem).

These assumptions support schemes such as:

- Kyber / ML-KEM,
- Dilithium / ML-DSA,
- NTRU,
- fully homomorphic encryption constructions.

Why Lattices Matter

Lattice-based cryptography is central because:

- it is currently the most influential practical PQC family,
- it supports both encryption/KEM and signature constructions,
- and it has strong connections between average-case and worst-case hardness in several settings.

From the attack perspective, the dominant language is:

- **lattice reduction** (LLL, BKZ),
- **sieving**,
- and careful **parameter estimation**.

Lattice Attack Perspective

According to the table, the best known attacks remain broadly **exponential** in the relevant dimension parameter.

This is the key orientation point:

- there is no Shor-style polynomial-time quantum algorithm known for the main lattice assumptions,
- but concrete security depends heavily on accurate cost models for BKZ and related reductions,
- so lattice security is a field where asymptotics and concrete cryptanalysis must be studied together.

Code-Based Problems

Code-based cryptography is centered on problems such as:

- **syndrome decoding**,
- more generally, the difficulty of decoding random linear codes.

Representative systems include:

- McEliece,
- HQC,
- BIKE.

Code-Based Attack Perspective

The dominant attack family is **Information Set Decoding (ISD)** and its variants.

The main qualitative lesson is:

- the best known attacks remain **exponential**,
- quantum speedups appear much more limited than in factorization or DLP,
- and Grover-type improvements generally give only a **quadratic speedup**, rather than a full structural collapse.

This makes code-based cryptography one of the most conservative PQC directions.

Hash-Based Assumptions

Hash-based cryptography relies on familiar assumptions:

- **collision resistance**,
- **preimage resistance**,
- **second-preimage resistance**.

Representative systems include:

- SPHINCS+,
- XMSS,
- LMS.

Hash-Based Security Perspective

The attack picture is comparatively clean:

- collisions face the **birthday bound**,
- preimage search is affected by **Grover's algorithm**,
- but there is no analogous Shor-style collapse.

Therefore, hash-based signatures are often viewed as:

- conceptually conservative,
- relatively robust at the assumption level,
- though sometimes less attractive in size or efficiency.

Multivariate Polynomial Problems

Multivariate cryptography is built on the difficulty of solving systems of quadratic equations over finite fields.

This is often referred to as the **MQ problem**.

Representative systems include:

- UOV,
- MAYO,
- historically, Rainbow.

Multivariate Attack Perspective

The table highlights the relevant attack language:

- **MinRank**-type attacks,
- **Kipnis–Shamir**-type methods,
- **XL / Gröbner basis** approaches,
- and scheme-specific structural attacks.

The critical lesson is that multivariate systems may appear hard in the abstract, yet particular constructions can fail because of exploitable algebraic structure.

Structural Fragility: The Rainbow Lesson

Rainbow is an important case study.

- At the high level, MQ remains a hard class of problems.
- But Rainbow itself was broken through a much more specific structural attack.
- This teaches that **scheme hardness** is not identical to **family hardness**.

This distinction will be a recurring theme throughout the colloquium.

Isogeny Problems

Isogeny-based cryptography emerged as a highly elegant and compact direction in PQC.

Systems discussed in this space include:

- SQISign,
- CSIDH,
- and historically SIKE / SIDH.

The Isogeny Cautionary Story

The table captures one of the most important cautionary episodes in modern PQC:

- **SIKE is broken**.
- The Castryck–Decru attack showed that a scheme can collapse completely even when the surrounding research area still appears mathematically rich.

More broadly:

- some isogeny problems still resist known attacks,
- but the family must now be approached with increased caution,
- especially when hidden structure may create special-purpose vulnerabilities.

A Comparative Reading of the Hard Problem Landscape

Based on the table, the families differ in a striking way:

- **Factorization / DLP / ECDLP**: strong classically, but broken by polynomial-time quantum algorithms.
- **Lattice / Code-based**: still believed hard; attacks remain exponential or subexponential only in restricted senses.
- **Hash-based**: quantum attacks are limited and well understood.
- **Multivariate / Isogeny**: mathematically interesting, but some constructions have shown serious fragility.

Three Evaluation Questions for Every Scheme

For each cryptographic construction studied in this colloquium, we should ask:

1. **What exact hard problem is being assumed?**
2. **What are the best known attacks, classically and quantumly?**
3. **Does the scheme introduce exploitable structure beyond the underlying problem?**

This framework helps separate genuine hardness from merely optimistic design.

What the Table Really Teaches

The information in the table should not be read as a static catalog.

It should be read as a set of research attitudes:

- compare assumptions by their **attack surface**,
- distinguish **family-level hardness** from **scheme-level security**,
- treat broken constructions as lessons, not merely failures,
- and always ask whether the best known attack is generic, structural, classical, or quantum.

How This Colloquium Can Be Structured

A natural progression for the colloquium is:

1. Classical hard problems and the quantum threat
2. Lattice assumptions and reduction-based security
3. Code-based cryptography and ISD
4. Hash-based signatures as a conservative baseline
5. Multivariate cryptography and structural attacks
6. Isogeny cryptography after SIKE
7. Comparative security evaluation across families

Expected Outcomes

By the end of the colloquium, participants should be able to:

- explain the difference between classical and post-quantum hardness assumptions,
- describe the principal attack models against each family,
- identify why some assumptions remain credible and others do not,
- and evaluate a PQC scheme in terms of both mathematics and cryptanalysis.

Closing Perspective

Final Perspective

Post-quantum cryptography is not merely a search for new schemes. It is a disciplined search for mathematical problems whose hardness survives new computational models, new attack paradigms, and deeper structural analysis.

That is the central intellectual purpose of the PQC Hard Problem Colloquium.

Thank you.

Discussion and questions